Beliefs, learning, and personality in the indefinitely repeated prisoner's dilemma

Web Appendix
(Intended for Online Publication)

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Web Appendix I

Screenshots from the experiment

Welcome!

Today's experiment will last about 60 minutes.

You will be paid a show-up fee of $5 together with any money you accumulate during this experiment. The amount of money you accumulate will depend partly on your actions and partly on the actions of other participants. This money will be paid at the end of the experiment in private and in cash.

It is important that during the experiment you remain **silent**. If you have a question or need assistance of any kind, please **raise your hand, but do not speak** - and an experiment administrator will come to you, and you may then whisper your question.

In addition, please **turn off your cell phones and put them away now**. Please do not use or place on your desk any personal items, including pens, paper, phones etc. Please do not look into anyone else's booth at any time.

Anybody that breaks these rules will be asked to leave.

Agenda:

- Questionnaire and test
- Experimental Instructions
- Experiment

Next

Figure A.1: Introductory screen

How Matches Work

The experiment is made up of **25 matches**.

At the start of each match you will be randomly paired with another participant in this room.

You will then play a number of rounds with that participant (this is what we call a "match").

Each match will last for a random number of **rounds**:

- At the end of each round the computer will roll a four-sided fair dice.
- If the computer rolls a 1, 2, or 3, then the match continues for at least one more round (75% probability).
- If the computer rolls a 4, then the match ends (25% probability).

When the match ends, you will again be randomly paired with another participant for the next match.

Choices and Payoffs

In each round of a match, you will choose A or B. The participant you are matched with will also choose A or B.

The table below describes the payoffs from the four pairs of choices that are possible in each round of a match.

These payoffs are in **points**.

My Choice in Round X	A	A	B	B
Other's Choice in Round X	A	B	A	B
My Payoff in Round X	32	12	50	25
Other's Payoff in Round X	32	50	12	25

In words:

- If you choose A, and the participant you are matched with chooses A, then your payoff for the round will be 32 points, and the other's payoff will be 32 points.
- If you choose A, and the participant you are matched with chooses B, then your payoff for the round will be 12 points, and the other's payoff will be 50 points.
- If you choose B, and the participant you are matched with chooses A, then your payoff for the round will be 50 points, and the other's payoff will be 12 points.
- If you choose B, and the participant you are matched with chooses B, then your payoff for the round will be 25 points, and the other's payoff will be 25 points.

At the end of the experiment, your total points will be converted into cash at the exchange rate of 250 points = $1. In addition, you will be paid your show-up fee of $5.

Next

Figure A.2: Description of matches, choices and payoffs

Practice Matches

To get some practice with the choices, you will now play 10 practice matches.

In these practice matches you are not yet matched with another participant and you will not earn any money.

In these practice matches you will make choices both for yourself and for the other participant.

Practice Match #2 of 10

The table below describes the payoffs from the four pairs of choices that are possible in each round of a match.

My Choice in Round X	A	A	B	B
Other's Choice in Round X	A	B	A	B
My Payoff in Round X	32	12	50	25
Other's Payoff in Round X	32	50	12	25

Remember that at the end of each round the computer rolls a four-sided fair dice. The match ends when the computer rolls a 4.

Dice Roll		2	1
Round	3	2	1
My Choice		B	A
Other's Choice		A	A
My Payoff		50	32
Other's Payoff		12	32

My Choice in Round **3**

A B

Other's Choice in Round **3**

A B

Figure A.3: Practice matches: Choice screen

Practice Matches

To get some practice with the choices, you will now play 10 practice matches.

In these practice matches you are not yet matched with another participant and you will not earn any money.

In these practice matches you will make choices both for yourself and for the other participant.

Practice Match #2 of 10

The table below describes the payoffs from the four pairs of choices that are possible in each round of a match.

My Choice in Round X	A	A	B	B
Other's Choice in Round X	A	B	A	B
My Payoff in Round X	32	12	50	25
Other's Payoff in Round X	32	50	12	25

Remember that at the end of each round the computer rolls a four-sided fair dice. The match ends when the computer rolls a 4.

Dice Roll	4	2	3	2	1
Round	5	4	3	2	1
My Choice	A	B	A	B	A
Other's Choice	B	B	A	A	A
My Payoff	12	25	32	50	32
Other's Payoff	50	25	32	12	32

Practice Match #2 Summary
Match duration: **5** rounds
My total payoff: **151** points
Other's total payoff: **151** points

Next

Figure A.4: Practice matches: Summary screen

Plans

When you are paired with another participant for a match, you will select a plan that will make your choices for you in every round. The other participant will also select a plan.

There are 10 possible plans.

Plans	
#1	Choose randomly between A and B in every round. At the beginning of every round, the computer flips a computerized fair coin for you: when your coin comes up heads, you choose A; when your coin comes up tails you choose B.
#2	Choose A in round 1. In round 2: choose A if the other chose A in round 1; choose B if the other chose B in round 1. After round 2: choose A if the other chose A in both of the previous two rounds; choose B if the other chose B in either of the previous two rounds.
#3	Choose B in round 1. After round 1: choose A if the other chose A in the previous round; choose B if the other chose B in the previous round.
#4	Choose B in every round.
#5	Choose A in round 1. After round 1: choose A if the other chose A in the previous round; choose B if the other chose B in the previous round.
#6	Choose A in round 1. After round 1: choose A if the other chose A in every one of the previous rounds; choose B if the other chose B in one or more of the previous rounds.
#7	Choose A in rounds 1 and 2. After round 2: choose A if the other has never chosen B twice in a row (i.e., if the other has never chosen B in two consecutive previous rounds); choose B if the other has ever chosen B twice in a row.
#8	Choose A in rounds 1 and 2. After round 2: choose A if the other chose A in either of the previous two rounds; choose B if the other chose B in both of the previous two rounds.
#9	Choose A in every round.
#10	Choose B in round 1. After round 1: choose A if the other chose A in every one of the previous rounds; choose B if the other chose B in one or more of the previous rounds.

Review Plans

Time Left: **2:42**

You have 3 minutes to review the table presenting the 10 plans.

You will have access to this table throughout the experiment.

Figure A.5: Review of strategies

Notes: The order that the ten strategies appeared on the subject's screen was randomized across subjects (see footnote 19). In this example, the order of the strategies is: RAND; 2TFT; DTFT; AD; TFT; G; G2; TF2T; AC; DG.

Plan Testing Stage

Time Left: **4:10**

You now have 5 minutes to test how plans make choices.

In this plan testing stage you are not yet matched with another participant and you will not earn any money.

Plans		My Plan	Other's Plan
#1	Choose randomly between A and B in every round. At the beginning of every round, the computer flips a computerized fair coin for you: when your coin comes up heads, you choose A; when your coin comes up tails you choose B.	Select	Select
#2	Choose A in round 1. In round 2: choose A if the other chose A in round 1; choose B if the other chose B in round 1. After round 2: choose A if the other chose A in both of the previous two rounds; choose B if the other chose B in either of the previous two rounds.	Select	Select
#3	Choose B in round 1. After round 1: choose A if the other chose A in the previous round; choose B if the other chose B in the previous round.	Select	Select
#4	Choose B in every round.	Select	Select
#5	Choose A in round 1. After round 1: choose A if the other chose A in the previous round; choose B if the other chose B in the previous round.	Select	Select
#6	Choose A in round 1. After round 1: choose A if the other chose A in every one of the previous rounds; choose B if the other chose B in one or more of the previous rounds.	Select	Select
#7	Choose A in rounds 1 and 2. After round 2: choose A if the other has never chosen B twice in a row (i.e., if the other has never chosen B in two consecutive previous rounds); choose B if the other has ever chosen B twice in a row.	Select	Select
#8	Choose A in rounds 1 and 2. After round 2: choose A if the other chose A in either of the previous two rounds; choose B if the other chose B in both of the previous two rounds.	Select	Select
#9	Choose A in every round.	Select	Select
#10	Choose B in round 1. After round 1: choose A if the other chose A in every one of the previous rounds; choose B if the other chose B in one or more of the previous rounds.	Select	Select

The table below describes the payoffs from the four pairs of choices that are possible in each round of a match.

My Choice in Round X	A	A	B	B
Other's Choice in Round X	A	B	A	B
My Payoff in Round X	32	12	50	25
Other's Payoff in Round X	32	50	12	25

Remember that at the end of each round the computer rolls a four-sided fair dice. The match ends when the computer rolls a 4.

Test Match My Plan: **Plan #2** Other's Plan: **Plan #10**

Dice Roll	4	3	1	2	1	2
Round	6	5	4	3	2	1
My Choice	B	B	B	B	B	A
Other's Choice	B	B	B	B	A	B
My Payoff	25	25	25	25	50	12
Other's Payoff	25	25	25	25	12	50

Test Match Summary

Match duration: **6** rounds

My total payoff: **162** points

Other's total payoff: **162** points

Figure A.6: Testing strategies: Summary screen

Reminders

1. The experiment is made up of 25 matches.

2. At the start of each match you will be randomly paired with another participant.

3. You will select a plan that will make your choices for you in every round of the match. The other participant will also select a plan.

4. The match will last a random number of rounds.

5. When the match ends, you will again be randomly paired with another participant for the next match.

6. At the end of the experiment, your total points will be converted into cash at the exchange rate of 200 points = $1. In addition, you will be paid your show-up fee of $5.

Begin Experiment

Figure A.7: Reminder screen

Match #1 of 25

You have been randomly paired with another participant for Match #1.

Please select your plan for Match #1 and then confirm your choice.

Plans		My Plan
#1	Choose randomly between A and B in every round. At the beginning of every round, the computer flips a computerized fair coin for you: when your coin comes up heads, you choose A; when your coin comes up tails you choose B.	Select
#2	Choose A in round 1. In round 2: choose A if the other chose A in round 1; choose B if the other chose B in round 1. After round 2: choose A if the other chose A in both of the previous two rounds; choose B if the other chose B in either of the previous two rounds.	Select
#3	Choose B in round 1. After round 1: choose A if the other chose A in the previous round; choose B if the other chose B in the previous round.	Select
#4	Choose B in every round.	Select
#5	Choose A in round 1. After round 1: choose A if the other chose A in the previous round; choose B if the other chose B in the previous round.	Select
#6	Choose A in round 1. After round 1: choose A if the other chose A in every one of the previous rounds; choose B if the other chose B in one or more of the previous rounds.	Select
#7	Choose A in rounds 1 and 2. After round 2: choose A if the other has never chosen B twice in a row (i.e., if the other has never chosen B in two consecutive previous rounds); choose B if the other has ever chosen B twice in a row.	Select
#8	Choose A in rounds 1 and 2. After round 2: choose A if the other chose A in either of the previous two rounds; choose B if the other chose B in both of the previous two rounds.	Select
#9	Choose A in every round.	Select
#10	Choose B in round 1. After round 1: choose A if the other chose A in every one of the previous rounds; choose B if the other chose B in one or more of the previous rounds.	Select

The table below describes the payoffs from the four pairs of choices that are possible in each round of a match.

My Choice in Round X	A	A	B	B
Other's Choice in Round X	A	B	A	B
My Payoff in Round X	32	12	50	25
Other's Payoff in Round X	32	50	12	25

Remember that at the end of each round the computer rolls a four-sided fair dice. The match ends when the computer rolls a 4.

My Plan: **--please select--** Confirm Choice

Dice Roll

Round

My Choice

Other's Choice

My Payoff

Other's Payoff

Figure A.8: Supergame 1: Choice screen

Question Before We Implement Your Plan in Match # 1

Before we implement your plan in Match #1, **on the next screen we will ask you what you think the chances are that the other participant chooses each of the ten plans.**

In other words, we will ask you how many times out of 100 you think the other participant in Match #1 would choose each of the ten plans.

Of course, the other participant chooses his/her plan for Match #1 only once (just like you), not 100 times. But you can think of the question as a way of asking how likely the other participant is to choose each of the plans.

You will be paid according to the accuracy of your answer. <u>You will make the most money on average if you answer truthfully what you think the chances are that the other participant chooses each of the plans.</u>

You will not be paid for your answer until the end of the experiment. Your answer will not be shown to any other participant. Your answer will not affect the experiment in any way.

You do not need to understand the details of how the payment works. If you are interested, the details follow below. If you are not interested, you can stop reading this screen now.

Details of how the payment works:

> *For this task, you start with 400 points and you will then be penalized for inaccuracy. The total penalty can never be more than 400 points. Your payment for this task will be 400 points minus the total penalty.*
>
> *For each of the ten possible plans, the penalty will be calculated as follows.*
>
> 1. *If that plan is chosen by the other participant, then the penalty will be smaller the higher your answer about the chances that the other participant chooses that plan. In that case, the penalty is calculated as follows:*
>
> $$(100 - \text{Your Answer}) \times (100 - \text{Your Answer}) \times (0.02 \text{ points})$$
>
> 2. *If that plan is not chosen by the other participant, then the penalty will be smaller the lower your answer about the chances that the other participant chooses that plan. In that case, the penalty is calculated as follows:*
>
> $$(\text{Your Answer}) \times (\text{Your Answer}) \times (0.02 \text{ points})$$

Next

Figure A.9: Supergame 1: Belief elicitation information

Question Before We Implement Your Plan in Match # 1

Plans		Other's Plan (Chance, %)
#1	Choose randomly between A and B in every round. At the beginning of every round, the computer flips a computerized fair coin for you: when your coin comes up heads, you choose A; when your coin comes up tails you choose B.	
#2	Choose A in round 1. In round 2: choose A if the other chose A in round 1; choose B if the other chose B in round 1. After round 2: choose A if the other chose A in both of the previous two rounds; choose B if the other chose B in either of the previous two rounds.	
#3	Choose B in round 1. After round 1: choose A if the other chose A in the previous round; choose B if the other chose B in the previous round.	
#4	Choose B in every round.	
#5	Choose A in round 1. After round 1: choose A if the other chose A in the previous round; choose B if the other chose B in the previous round.	
#6	Choose A in round 1. After round 1: choose A if the other chose A in every one of the previous rounds; choose B if the other chose B in one or more of the previous rounds.	
#7	Choose A in rounds 1 and 2. After round 2: choose A if the other has never chosen B twice in a row (i.e., if the other has never chosen B in two consecutive previous rounds); choose B if the other has ever chosen B twice in a row.	
#8	Choose A in rounds 1 and 2. After round 2: choose A if the other chose A in either of the previous two rounds; choose B if the other chose B in both of the previous two rounds.	
#9	Choose A in every round.	
#10	Choose B in round 1. After round 1: choose A if the other chose A in every one of the previous rounds; choose B if the other chose B in one or more of the previous rounds.	
	The sum must add up to 100: Check Sum	

What do you think the chances are that the other participant chooses each of the ten plans?

In other words, how many times out of 100 do you think the other participant in Match #1 would choose each of the ten plans?

- Please use only whole numbers for your answer.
- You will be paid according to the accuracy of your answer. You will make the most money on average if you answer truthfully what you think the chances are that the other participant chooses each of the plans.
- You will not be paid for your answer until the end of the experiment.
- Your answer will not be shown to any other participant.
- Your answer will not affect the experiment in any way.

Figure A.10: Supergame 1: Belief elicitation

Match #1 of 25

Plans		My Plan
#1	Choose randomly between A and B in every round. At the beginning of every round, the computer flips a computerized fair coin for you: when your coin comes up heads, you choose A; when your coin comes up tails you choose B.	Select
#2	Choose A in round 1. In round 2: choose A if the other chose A in round 1; choose B if the other chose B in round 1. After round 2: choose A if the other chose A in both of the previous two rounds; choose B if the other chose B in either of the previous two rounds.	Select
#3	Choose B in round 1. After round 1: choose A if the other chose A in the previous round; choose B if the other chose B in the previous round.	Select
#4	Choose B in every round.	Select
#5	Choose A in round 1. After round 1: choose A if the other chose A in the previous round; choose B if the other chose B in the previous round.	Select
#6	Choose A in round 1. After round 1: choose A if the other chose A in every one of the previous rounds; choose B if the other chose B in one or more of the previous rounds.	Select
#7	Choose A in rounds 1 and 2. After round 2: choose A if the other has never chosen B twice in a row (i.e., if the other has never chosen B in two consecutive previous rounds); choose B if the other has ever chosen B twice in a row.	Select
#8	Choose A in rounds 1 and 2. After round 2: choose A if the other chose A in either of the previous two rounds; choose B if the other chose B in both of the previous two rounds.	Select
#9	Choose A in every round.	Select
#10	Choose B in round 1. After round 1: choose A if the other chose A in every one of the previous rounds; choose B if the other chose B in one or more of the previous rounds.	Select

The table below describes the payoffs from the four pairs of choices that are possible in each round of a match.

My Choice in Round X	A	A	B	B
Other's Choice in Round X	A	B	A	B
My Payoff in Round X	32	12	50	25
Other's Payoff in Round X	32	50	12	25

Remember that at the end of each round the computer rolls a four-sided fair dice. The match ends when the computer rolls a 4.

My Plan: **Plan #4**

Dice Roll	4	1	3	2
Round	4	3	2	1
My Choice	B	B	B	B
Other's Choice	B	B	A	A
My Payoff	25	25	50	50
Other's Payoff	25	25	12	12

Match # 1 Summary

Match duration: **4** rounds

My total payoff: **150** points

Other's total payoff: **74** points

Next

Figure A.11: Supergame 1: Summary screen

Web Appendix II
Details referred to in footnotes

Web Appendix II.1 Details referred to in footnote 5

Li and Liu (2017) find a weak relationship between group identity and beliefs in the first round of their first indefinitely repeated prisoner's dilemma game. Dreber et al. (2014) find that cooperation in indefinitely repeated prisoner's dilemma games with mistakes in implementation is correlated with beliefs (measured after the games) about the likelihood that defections were due to mistakes. When subjects play a sequence of indefinitely repeated prisoner's dilemma games where they are randomly rematched after every round within a supergame: (i) Duffy and Ochs (2009) find that cooperation is low and can be predicted well using threshold strategies elicited in an earlier one-shot prisoner's dilemma together with subjects' round-by-round forecasts of how many others in the matching group of fourteen will cooperate in that round; and (ii) Duffy and Fehr (2018) elicit beliefs about how many others in the matching group of ten will cooperate in the first round of each supergame, finding that beliefs and behavior are correlated and that beliefs respond to experience in an earlier stag hunt game.

Web Appendix II.2 Details referred to in footnote 6

In one-shot prisoner's dilemmas, see: Messé and Sivacek (1979); Croson (2000); Miettinen and Suetens (2008); Charness et al. (2016); Engel and Zhurakhovska (2016); Ridinger and McBride (2017); Peeters and Vorsatz (2018); Heuer and Orland (2019); and Sutter and Untertrifaller (2020). Croson (2000) also finds that eliciting beliefs lowers cooperation (but see footnote 8 in the main text), while Charness et al. (2016) also find that beliefs change with the payoff from joint cooperation. In Miettinen and Suetens (2008), beliefs are more cooperative when both players send a message expressing a desire for mutual cooperation.

Web Appendix II.3 Details referred to in footnote 9

In one-shot or finitely repeated prisoner's dilemmas, Haesevoets et al. (2018) find that a survey measure of trust predicts cooperation, Emonds et al. (2014) find that a survey measure of trust predicts cooperation among prosocial subjects, while Acedo-Carmona and Gomila (2014) find that subjects cooperate more when they are matched with people they know and trust personally, although Ahn et al. (2003) find no effect of a survey measure of trust. Papers that correlate survey measures of trust with contributions in one-shot or finitely repeated linear public goods games mostly find a positive effect of trust (Sato, 1988, 1989; Anderson et al., 2004; Gächter et al., 2004; Thöni et al., 2012; Peysakhovich et al., 2014), while Mulder et al. (2006) find

that experiencing a treatment with sanctions undermines contributions of high trust subjects. Finally, in a real-effort game, Proto and Rustichini (2014) find that trust predicts whether subjects choose effort consistent with believing that others are cooperative.

Web Appendix II.4 Details referred to in footnote 36

The anxiety, cautiousness and trust measures include ten questions each and, as described above, are facets of the Big Five; the questions come from the 300-item IPIP-NEO (see Goldberg, 1999, and Johnson, 2014) and are at: `https://ipip.ori.org/newNEOFacetsKey.htm`. The manipulativeness measure includes six questions and is one of thirty-three scales from the Computerized Adaptive Test of Personality Disorder (CAT–PD); the questions come from the 212-item CAT–PD–SF (see Simms et al., 2011, and Wright and Simms, 2014) and are at: `https://ipip.ori.org/newCAT-PD-SFv1.1Keys.htm`. The forgiveness and kindness measures include eight question each and are two of the twenty-four character strengths from the Values in Action Inventory of Strengths (VIA-IS); the questions come from the 192-item VIA-IS-R (see Peterson and Seligman, 2004, and McGrath, 2017) and for research purposes are available on request from the VIA Institute on Character (`https://www.viacharacter.org`).

Web Appendix II.5 Details referred to in footnote 38

All 52 questions use a five-point Likert scale. For consistency, we presented all questions in the form 'I ...', and we used the introductory wording and response categories recommended by IPIP at: `https://ipip.ori.org/New_IPIP-50-item-scale.htm`. The 52 questions were split into five screens of ten questions and a final screen of two questions. Subjects could change their answers on a particular screen until they submitted their answers for that screen. Subjects were allowed to submit incomplete sets of answers, but were asked to confirm that they wanted to do so. We replaced missing responses by the sample average of nonmissing responses to that particular question.

Web Appendix II.6 Details referred to in footnote 39

The ten highest loading questions underlying the trust factor are the ten questions that measure trust. The nine highest loading questions underlying the anxiety (cautiousness) factor come from the ten questions that measure anxiety (cautiousness). The six highest loading questions underlying the kindness factor come from the eight questions that measure kindness. The four highest loading questions underlying the manipulativeness factor come from the six questions that measure manipulativeness (and the other two questions also appear among the ten highest loading questions); the raw factor loads negatively on manipulativeness, and so to create the manipulativeness factor we changed the sign of all the factor loadings.

Web Appendix II.7 Details referred to in footnote 44

We ran a linear OLS regression of accuracy in Supergame 1 on the treatment, controlling for the five personality factors, demographic characteristics and standardized cognitive ability (see Section 2.9), and using heteroskedasticity-robust standard errors and two-sided tests of significance. The positive effect of $R = 48$ relative to $R = 32$ is significant at the one-percent level.

Web Appendix II.8 Details referred to in footnote 47

We ran the regression described in Web Appendix II.7, replacing accuracy as the dependent variable with an indicator taking value 1 if a subject chose a good response to her beliefs in Supergame 1, and further including optimism (defined in the notes to Figure 4) and the interaction of optimism with the treatment. Setting $R = 32$ ($R = 48$) as the omitted category, we find a negative (positive) effect of optimism on the probability of good responding, statistically significant at the one-percent (one-percent) level. We also find that the effect of optimism when $R = 48$ is significantly different to that when $R = 32$, again at the one-percent level.

Web Appendix III
Strategy transitions

In Table A.1 we study transitions from one strategy category to another (Figure 2 defines the strategy categories). Panel I uses the observations where the subject chose a strategy from the unfriendly category in the previous supergame, and we consider the factors that drive the subject to continue choosing an unfriendly strategy in the next supergame, to change to a provocable strategy, or to change to a lenient strategy. Panel II (III) repeats the exercise, using observations where the subject chose a strategy from the provocable (lenient) category in the previous supergame. The regressions reported in Table A.1 include the same independent variables as the regressions reported in Table 6 (to save space we do not report the treatment coefficients).

The first row of Panels I-III of Table A.1 show that when the previous supergame was longer, subjects are less likely to stay unfriendly (first column of Panel I), and more likely to stay provocable and lenient (second column of Panel II and third column of Panel III). Furthermore, the increased likelihood of staying provocable comes at the expense of changes from provocable to unfriendly strategies (first and second columns of Panel II). These findings help us to understand why previous supergame length has no effect on the likelihood of changing strategy (Table 6), while at the same time increasing cooperation (Table 5).

The second row of Panels I-III of Table A.1 show that when a subject's opponent cooperated more in the previous supergame, the subject is more likely to stay within the unfriendly, provocable and lenient categories (first column of Panel I, second column of Panel II, and third column of Panel III). Even though subjects are somewhat more likely to stay unfriendly, they are also much less likely to change from provocable or lenient strategies to unfriendly ones when their opponent in the previous supergame cooperated more (first columns of Panels II and III). These findings shed light on the mechanism by which the opponent's cooperation in the previous supergame reduces the likelihood of changing strategy (Table 6), while at the same time increasing the subject's own cooperation (Table 5).

Although the effects are not individually statistically significant, the fifth and six rows of Panels I-III of Table A.1 show that quality of thinking and the quality of the strategy chosen in the previous supergame reduce the likelihood of staying unfriendly (first column of Panel I), but increase the likelihood of staying provocable and lenient by more (second column of Panel II and third column of Panel III). These findings help to explain how quality of thinking and the quality of the strategy chosen in the previous supergame reduce the likelihood of changing strategy (Table 6). Finally, the seventh row shows that as subjects gain experience by playing more supergames, they are more likely to stay unfriendly, provocable and lenient; thus, the finding that subjects who have played more supergames change strategy less frequently (Table 6) holds at the level of all three strategy categories.

I: Unfriendly strategy in Supergame $t-1$

	(1) Stay unfriendly	(2) Change to provocable	(3) Change to lenient
Length of Sup. $t-1$	-0.006***	0.005***	0.002*
	(0.001)	(0.002)	(0.001)
Other's strategy coop in Sup. $t-1$	0.068***	-0.025*	-0.025**
	(0.017)	(0.014)	(0.011)
Own strategy coop in Sup. 1	-0.155***	0.109***	0.050*
	(0.040)	(0.026)	(0.027)
Own optimism in Sup. 1	-0.161***	0.048	0.100***
	(0.054)	(0.037)	(0.027)
Good responder to beliefs in Sup. 1	-0.004	-0.007	0.020
	(0.022)	(0.017)	(0.013)
Quality of Sup. $t-1$ strategy	-0.010	-0.025	0.006
	(0.084)	(0.070)	(0.037)
Supergame number	0.004***	-0.002**	-0.001***
	(0.001)	(0.001)	(0.000)
Mean of DV	0.827	0.102	0.045
N	4820	4820	4820

II: Provocable strategy in Supergame $t-1$

	(1) Change to unfriendly	(2) Stay provocable	(3) Change to lenient
Length of Sup. $t-1$	-0.009***	0.009***	-0.001
	(0.002)	(0.003)	(0.001)
Other's strategy coop in Sup. $t-1$	-0.236***	0.250***	-0.006
	(0.031)	(0.031)	(0.013)
Own strategy coop in Sup. 1	-0.175***	0.171***	0.023
	(0.040)	(0.058)	(0.022)
Own optimism in Sup. 1	-0.111*	0.079	0.029
	(0.057)	(0.073)	(0.029)
Good responder to beliefs in Sup. 1	-0.023	0.027	0.001
	(0.019)	(0.024)	(0.009)
Quality of Sup. $t-1$ strategy	-0.124	0.128	-0.020
	(0.076)	(0.087)	(0.026)
Supergame number	-0.003***	0.005***	-0.002***
	(0.001)	(0.001)	(0.001)
Mean of DV	0.185	0.746	0.052
N	2676	2676	2676

III: Lenient strategy in Supergame $t-1$

	(1) Change to unfriendly	(2) Change to provocable	(3) Stay lenient
Length of Sup. $t-1$	-0.002	-0.003	0.006*
	(0.003)	(0.002)	(0.003)
Other's strategy coop in Sup. $t-1$	-0.202***	-0.047	0.260***
	(0.046)	(0.032)	(0.042)
Own strategy coop in Sup. 1	-0.190***	-0.047	0.250***
	(0.049)	(0.049)	(0.076)
Own optimism in Sup. 1	-0.175	-0.004	0.183
	(0.114)	(0.092)	(0.151)
Good responder to beliefs in Sup. 1	-0.034	0.007	0.029
	(0.027)	(0.022)	(0.035)
Quality of Sup. $t-1$ strategy	-0.010	-0.117**	0.135
	(0.063)	(0.048)	(0.084)
Supergame number	-0.003*	-0.004***	0.006***
	(0.001)	(0.001)	(0.002)
Mean of DV	0.167	0.090	0.720
N	1495	1495	1495

Table A.1: Strategy category transitions from Supergame $t-1$ to Supergame t

Notes: Panel I uses all observations from Supergames 1 to 24 where the subject chose a strategy from the unfriendly category (Figure 2 defines the strategy categories). The first column of Panel I reports a linear OLS regression of a binary variable that takes value 1 if the subject continued to choose a strategy from the unfriendly category in the next supergame, and taking value 0 otherwise; note that the variable takes value 1 even if the subject changed strategy within the unfriendly category. The second (third) column of Panel I reports a linear OLS regression of a binary variable that takes value 1 if the subject changed her strategy to one from the provocable (lenient) category in the next supergame, and taking value 0 otherwise. Panels II and III are constructed similarly. Transition probabilities do not sum to one because subjects can change to RAND, which is not included in the unfriendly, provocable or lenient categories. All regressions control for the five personality factors, demographic characteristics and standardized cognitive ability (see Section 2.9), and the treatment (treatment coefficients are omitted here to save space). The independent variables are the same as those in Table 6; see the notes to that table. Heteroskedasticity-robust standard errors with clustering at the session level are shown in parentheses. ***, ** and * denote significance at the 1 percent, 5 percent and 10 percent levels (two-sided tests).

Web Appendix IV

Additional figures

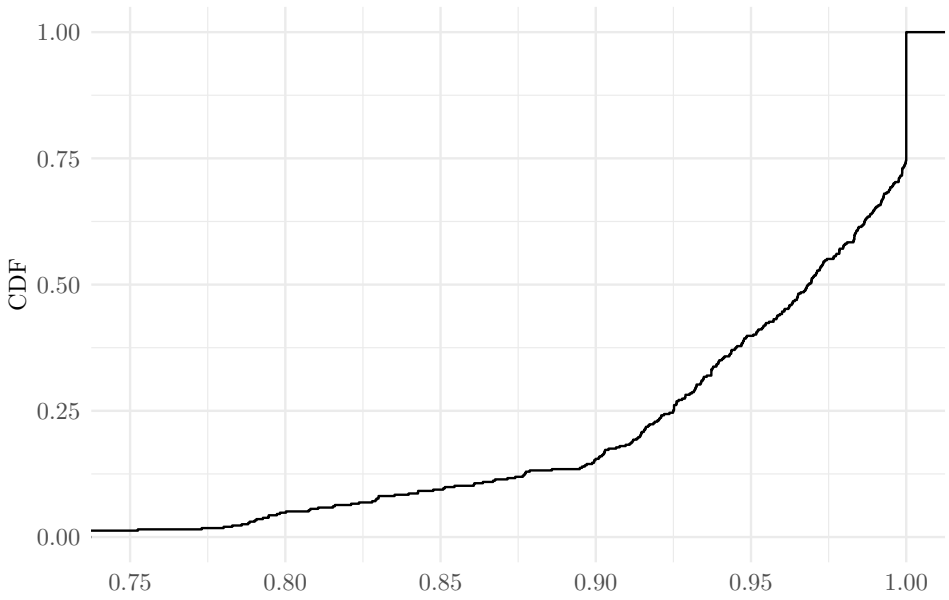

Figure A.12: CDF of payoffs relative to best responding to beliefs in Supergame 1

Notes: The figure shows the cumulative distribution function across subjects of the expected payoff of the subject's chosen strategy given her beliefs as a proportion of the expected payoff of the best response to her beliefs.

Web Appendix V

Additional tables

	AD	*DG*	*DTFT*	*RAND*	*G*	*2TFT*	*TFT*	*G2*	*TF2T*	*AC*
AD	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
DG	0.0000	0.0000	0.0000	0.1500	0.1875	0.1875	0.1875	0.7500	0.7500	0.7500
DTFT	0.0000	0.0000	0.0000	0.3750	0.1875	0.1875	0.4286	0.7500	0.7500	0.7500
RAND	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000
G	0.2500	0.2500	0.2500	0.4000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2TFT	0.2500	0.2500	0.2500	0.4844	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
TFT	0.2500	0.3906	0.5714	0.6250	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
G2	0.4375	1.0000	1.0000	0.7097	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
TF2T	0.4375	1.0000	1.0000	0.8594	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
AC	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table A.2: Cooperation rates

Notes: Each cell reports the cooperation rate of the strategy on the vertical axis when playing against the strategy on the horizontal axis. Cooperation rates measure the expected number of rounds of cooperation divided by the expected number of rounds, and are based on analytical calculations.

R	*AD*	*DG*	*DTFT*	*RAND*	*G*	*2TFT*	*TFT*	*G2*	*TF2T*	*AC*
32	0.91	0.02	0.05	0.01	0.01	0.01	0.02	0.02	0.02	0.02
40	0.23	0.08	0.04	0.00	0.05	0.01	0.01	0.67	0.15	0.05
48	0.04	0.03	0.01	0.00	0.10	0.04	0.05	0.85	0.18	0.04

Table A.3: Frequency each strategy is a best response in Supergame 1

Notes: This table replicates Table 2, but using best responding instead of good responding.

	R	*AD*	*DG*	*DTFT*	*RAND*	*G*	*2TFT*	*TFT*	*G2*	*TF2T*	*AC*
I: All	32	0.52	0.82	0.78	0.50	0.78	0.76	0.88	0.71	0.75	0.67
	40	0.76	1.00	1.00	1.00	0.71	0.72	0.70	0.55	0.55	0.61
	48	0.56	0.42	0.42	0.00	0.67	0.68	0.71	0.41	0.41	0.51
II: Pessimists	32	0.80	0.88	0.86	–	0.62	0.62	0.86	0.50	0.50	0.33
	40	0.84	1.00	1.00	–	0.62	0.64	0.57	0.34	0.34	0.12
	48	1.00	0.71	0.71	–	0.41	0.43	0.47	0.22	0.22	0.16
III: Optimists	32	0.26	0.71	0.67	0.50	0.90	0.89	0.89	0.88	0.88	1.00
	40	0.50	1.00	1.00	1.00	0.77	0.76	0.76	0.68	0.68	0.69
	48	0.00	0.00	0.00	0.00	0.91	0.91	0.91	0.60	0.60	0.61

Table A.4: Probability of good responding in Supergame 1
(conditional on each strategy being a good response)

Notes: For each strategy, the table shows the proportion of subjects who good respond to their beliefs, conditional on that strategy being in the set of good responses to the subject's beliefs. Good responding is defined in the second paragraph of Section 3.2.1. Panels II and III split subjects according to whether they are above or below the median level of optimism within-treatment, where optimism is defined in the notes to Figure 4 (subjects at the median are allocated to the above-median category).

	AD	*DG*	*DTFT*	*RAND*	*G*	*2TFT*	*TFT*	*G2*	*TF2T*	*AC*
AD	100.00	100.00	100.00	150.00	125.00	125.00	125.00	143.75	143.75	200.00
DG	100.00	100.00	100.00	140.70	115.25	115.25	129.31	146.00	146.00	146.00
DTFT	100.00	100.00	100.00	126.75	115.25	115.25	134.86	146.00	146.00	146.00
RAND	74.00	87.50	107.75	119.00	110.00	117.59	130.25	137.87	151.34	164.00
G	87.00	105.75	105.75	125.20	128.00	128.00	128.00	128.00	128.00	128.00
2TFT	87.00	105.75	105.75	119.97	128.00	128.00	128.00	128.00	128.00	128.00
TFT	87.00	98.44	113.14	111.25	128.00	128.00	128.00	128.00	128.00	128.00
G2	77.25	108.00	108.00	106.00	128.00	128.00	128.00	128.00	128.00	128.00
TF2T	77.25	108.00	108.00	96.72	128.00	128.00	128.00	128.00	128.00	128.00
AC	48.00	108.00	108.00	88.00	128.00	128.00	128.00	128.00	128.00	128.00

Table A.5: Payoffs when $R = 32$

Notes: Each cell reports the expected payoff in points of the strategy on the vertical axis when playing against the strategy on the horizontal axis. Expected payoffs are based on analytical calculations.

	AD	*DG*	*DTFT*	*RAND*	*G*	*2TFT*	*TFT*	*G2*	*TF2T*	*AC*
AD	100.00	100.00	100.00	150.00	125.00	125.00	125.00	143.75	143.75	200.00
DG	100.00	100.00	100.00	143.10	115.25	115.25	129.31	170.00	170.00	170.00
DTFT	100.00	100.00	100.00	132.75	115.25	115.25	134.86	170.00	170.00	170.00
RAND	74.00	89.90	113.75	127.00	116.40	125.34	140.25	149.23	165.09	180.00
G	87.00	105.75	105.75	131.60	160.00	160.00	160.00	160.00	160.00	160.00
2TFT	87.00	105.75	105.75	127.72	160.00	160.00	160.00	160.00	160.00	160.00
TFT	87.00	98.44	113.14	121.25	160.00	160.00	160.00	160.00	160.00	160.00
G2	77.25	132.00	132.00	117.35	160.00	160.00	160.00	160.00	160.00	160.00
TF2T	77.25	132.00	132.00	110.47	160.00	160.00	160.00	160.00	160.00	160.00
AC	48.00	132.00	132.00	104.00	160.00	160.00	160.00	160.00	160.00	160.00

Table A.6: Payoffs when $R = 40$

Notes: Each cell reports the expected payoff in points of the strategy on the vertical axis when playing against the strategy on the horizontal axis. Expected payoffs are based on analytical calculations.

	AD	*DG*	*DTFT*	*RAND*	*G*	*2TFT*	*TFT*	*G2*	*TF2T*	*AC*
AD	100.00	100.00	100.00	150.00	125.00	125.00	125.00	143.75	143.75	200.00
DG	100.00	100.00	100.00	145.50	115.25	115.25	129.31	194.00	194.00	194.00
DTFT	100.00	100.00	100.00	138.75	115.25	115.25	134.86	194.00	194.00	194.00
RAND	74.00	92.30	119.75	135.00	122.80	133.09	150.25	160.58	178.84	196.00
G	87.00	105.75	105.75	138.00	192.00	192.00	192.00	192.00	192.00	192.00
2TFT	87.00	105.75	105.75	135.47	192.00	192.00	192.00	192.00	192.00	192.00
TFT	87.00	98.44	113.14	131.25	192.00	192.00	192.00	192.00	192.00	192.00
G2	77.25	156.00	156.00	128.71	192.00	192.00	192.00	192.00	192.00	192.00
TF2T	77.25	156.00	156.00	124.22	192.00	192.00	192.00	192.00	192.00	192.00
AC	48.00	156.00	156.00	120.00	192.00	192.00	192.00	192.00	192.00	192.00

Table A.7: Payoffs when $R = 48$

Notes: Each cell reports the expected payoff in points of the strategy on the vertical axis when playing against the strategy on the horizontal axis. Expected payoffs are based on analytical calculations.

	(1)	(2)
R40	12.79***	12.58***
	(0.49)	(0.52)
R48	48.05***	46.99***
	(0.84)	(0.94)
Accuracy of beliefs		10.01***
		(2.71)
Proximity to best response	88.49***	87.14***
	(7.20)	(7.43)
Mean of dependent variable	128.15	128.15
N	390	390

Table A.8: Expected earnings in Supergame 1 (robustness)

Notes: The regressions reported here are the same as those reported in Columns 2 and 3 of Table 3 except that we replace 'Good responder to beliefs' with 'Proximity to best response', which measures the expected payoff of a subject's chosen strategy given her beliefs as a proportion of the expected payoff of the best response to her beliefs. To help interpret the effect size, note that our measure of proximity to the best response is defined from 0 to 1. Figure A.12 in Web Appendix IV shows the cumulative distribution function of proximity.

	(1)	(2)	(3)	(4)	(5)
R40	0.148***	0.135***	0.064	0.077	0.035
	(0.056)	(0.051)	(0.046)	(0.053)	(0.042)
R48	0.300***	0.275***	0.154***	0.199***	0.111***
	(0.042)	(0.038)	(0.043)	(0.045)	(0.038)
Length of Supergame $t-1$	0.007***				0.006***
	(0.001)				(0.001)
Other's Round 1 coop		0.087***			0.079***
in Supergame $t-1$		(0.019)			(0.016)
Own Round 1 coop			0.385***		0.286***
in Supergame 1			(0.036)		(0.034)
Own optimism				0.691***	0.353***
in Supergame 1				(0.073)	(0.064)
Mean of dependent variable	0.464	0.464	0.464	0.464	0.464
N	9360	9360	9360	9360	9360

Table A.9: Round 1 cooperation in Supergame t (robustness)

Notes: This table replicates Table 4, except that we use Probit regressions instead of linear OLS regressions. The table reports average marginal effects and heteroskedasticity-robust standard errors with clustering at the session level.

I: Strategy cooperation in Supergame t

	(1)	(2)	(3)	(4)	(5)
R40	0.103**	0.094**	0.059*	0.049	0.037
	(0.041)	(0.038)	(0.032)	(0.037)	(0.029)
R48	0.209***	0.191***	0.090***	0.128***	0.065**
	(0.033)	(0.030)	(0.029)	(0.029)	(0.025)
Length of Supergame $t-1$	0.004***				0.004***
	(0.001)				(0.001)
Other's inferred coop in Supergame $t-1$		0.095***			0.088***
		(0.021)			(0.017)
Own strategy coop in Supergame 1			0.422***		0.333***
			(0.034)		(0.036)
Own optimism in Supergame 1				0.556***	0.232***
				(0.048)	(0.039)
Mean of dependent variable	0.467	0.467	0.467	0.467	0.467
N	9360	9360	9360	9360	9360

II: Optimism in Supergame 25

	(1)	(2)	(3)	(4)	(5)
R40	0.110**	0.011	0.086*	0.062	-0.028
	(0.045)	(0.023)	(0.042)	(0.043)	(0.023)
R48	0.244***	0.049	0.178***	0.171***	-0.014
	(0.043)	(0.033)	(0.045)	(0.041)	(0.033)
Length of Supergames 1 to 24	0.040				0.024**
	(0.024)				(0.012)
Others' inferred coop in Supergames 1 to 24		1.027***			0.947***
		(0.122)			(0.110)
Own strategy coop in Supergame 1			0.232***		0.061*
			(0.036)		(0.033)
Own optimism in Supergame 1				0.493***	0.419***
				(0.042)	(0.042)
Mean of dependent variable	0.430	0.430	0.430	0.430	0.430
N	390	390	390	390	390

Table A.10: Effect of experience on behavior and beliefs (robustness)

Notes: The regressions reported in Panel I are the same as those reported in Panel I of Table 5, except that we replace 'Other's strategy coop in Supergame $t-1$' with 'Other's inferred coop in Supergame $t-1$'. The regressions reported in Panel II are the same as those reported in Panel II of Table 5, except that we replace 'Others' strategy coop in Supergames 1 to 24' with 'Others' inferred coop in Supergames 1 to 24'. 'Other's inferred coop in Supergame t' measures the weighted average of the strategy cooperation of each of the ten strategies, where the weights come from the posterior distribution over the opponent's strategies after Bayesian updating from the realized sequence of play in Supergame t; the Bayesian update for Supergame t uses the uniform distribution as the prior, and so the Bayesian update for Supergame t does not use information from sequences of play in prior supergames. See the notes to Table 5 for the definition of strategy cooperation. 'Others' inferred coop in Supergames 1 to 24' is the mean of 'Other's inferred coop in Supergame t' over the first 24 supergames.

	(1) Strategy cooperation (Supergames 2-25)	(2) Strategy cooperation (Supergames 21-25)	(3) Optimism (Supergame 25)
Anxiety	-0.004	-0.023*	-0.005
	(0.009)	(0.013)	(0.012)
Cautiousness	-0.009	-0.011	-0.003
	(0.011)	(0.011)	(0.015)
Kindness	-0.004	-0.010	0.009
	(0.014)	(0.017)	(0.009)
Manipulativeness	-0.009	-0.005	-0.013
	(0.013)	(0.015)	(0.013)
Trust	0.021*	0.022**	0.024**
	(0.010)	(0.008)	(0.010)
Mean of dependent variable	0.467	0.461	0.430
N	9360	1950	390
Control for beliefs in Sup. 1	No	No	No
Control for behavior in Sup. 1	No	No	No
Controls for experience	No	No	No

Table A.11: Effect of personality on behavior and beliefs (robustness)

Notes: The regressions reported here are the same as those reported in Table 7, except that they exclude the controls for experience and for Supergame 1 behavior and beliefs presented in rows three to six of Panel I of Table 5 (in relation to Columns 1 and 2 here) and of Panel II of Table 5 (in relation to Column 3 here); Columns 1 and 2 also exclude the supergame number control (not relevant to Column 3).

	(1) Strategy cooperation (Supergame 1)	(2) Strategy cooperation (Supergame 1)	(3) Optimism (Supergame 1)
Anxiety	0.009 (0.019)	0.017 (0.016)	-0.008 (0.011)
Cautiousness	-0.002 (0.017)	0.000 (0.013)	-0.002 (0.012)
Kindness	-0.022 (0.017)	-0.006 (0.014)	-0.017 (0.011)
Manipulativeness	-0.027 (0.019)	-0.008 (0.015)	-0.020* (0.012)
Trust	-0.005 (0.018)	-0.010 (0.013)	0.005 (0.012)
Mean of dependent variable	0.478	0.478	0.536
N	390	390	390
Control for beliefs in Sup. 1	No	Yes	—
Control for behavior in Sup. 1	—	—	No
Controls for experience	—	—	—

Table A.12: Effect of personality on behavior and beliefs in Supergame 1

Column 1 (Column 3) reports a linear OLS regression of strategy cooperation (optimism) in Supergame 1 on the five personality factors, controlling for demographic characteristics and standardized cognitive ability (see Section 2.9), and the treatment. The regression reported in Column 2 is the same as the one reported in Column 1, except that it further includes optimism in Supergame 1 as a control. 'Strategy cooperation' is defined in the notes to Table 5 and 'optimism' is defined in the notes to Figure 4. $N = 390$ because four subjects did not complete the demographic questionnaire. Heteroskedasticity-robust standard errors are shown in parentheses. ***, ** and * denote significance at the 1 percent, 5 percent and 10 percent levels (two-sided tests).

	(1)	(2)	(3)
R40	0.039	0.039	0.039
	(0.029)	(0.030)	(0.029)
R48	0.068**	0.060**	0.060**
	(0.026)	(0.025)	(0.025)
Length of Supergame $t-1$	0.004***	0.004***	0.004***
	(0.001)	(0.001)	(0.001)
Own strategy coop	0.334***	0.334***	0.334***
in Supergame 1	(0.036)	(0.036)	(0.036)
Own optimism	0.232***	0.231***	0.231***
in Supergame 1	(0.039)	(0.039)	(0.039)
Other's strategy coop	0.066***	0.085***	0.082***
in Supergame $t-1$	(0.014)	(0.023)	(0.024)
('Other's strategy coop		-0.037	-0.035
in Sup. $t-1$' $- x_R)_+$		(0.039)	(0.040)
Trust $\times$ Other's strategy			-0.054**
coop in Supergame $t-1$			(0.020)
Trust $\times$ ('Other's strategy			0.079**
coop in Sup. $t-1$' $- x_R)_+$			(0.037)
Mean of dependent variable	0.467	0.467	0.467
N	9360	9360	9360

Table A.13: Strategy cooperation in Supergame t

Notes: The regression reported in Column 1 is exactly the same as the one reported in Column 5 of Panel I of Table 5. The notes to Figure 8 describe how we run a piece-wise linear spline regression by further including $\max\{0, (\text{'Other's strategy coop in Supergame } t-1\text{'} - x_R)\}$, which for conciseness we label here as ('Other's strategy coop in Sup. $t-1$' $- x_R)_+$. Column 2 reports coefficients from this spline regression without interactions with trust, while Column 3 reports coefficients with interactions with trust. All regressions control for the five personality factors (including trust), demographic characteristics and standardized cognitive ability (see Section 2.9), and the supergame number, with $R = 32$ as the omitted category. N is in multiples of 390 because four subjects did not complete the demographic questionnaire. Heteroskedasticity-robust standard errors with clustering at the session level are shown in parentheses. ***, ** and * denote significance at the 1 percent, 5 percent and 10 percent levels (two-sided tests).

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